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Handwritten Arabic Text Recognition: A Review of Recent Trends and Future Direction

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Abstract

Handwriting recognition is an important research area in the domains of computer vision (CV), pattern recognition, and Natural Language Processing (NLP). Accurate recognition of handwritten text has numerous applications, including digitizing historical documents, automating data entry, and document analysis or verification. Handwriting recognition has been researched widely and applied to the scripts of various languages including Latin, Chinese, Hindi, and Arabic. This study focuses on offline handwritten Arabic text recognition (HATR), addressing the unique challenges posed by Arabic, which is a language spoken widely across the globe. Handwritten Arabic text poses significant challenges in handwriting recognition because of its cursive nature, rich calligraphic styles and varied diacritics. Hence, various methods have been explored to address these challenges and complexities. This paper presents a comprehensive study of the research advances in offline HATR. It includes review of the background, challenges, the strengths of methods proposed in the literature, the solutions offered, as well as highlights the weaknesses of each technique and finally, future research direction.

Keywords Arabic handwriting · Offline Handwritten Arabic text recognition · Computer vision · Deep learning · Diacritics · Offline recognition · Optical character recognition

1 Introduction

Optical Character Recognition (OCR) is defined as the digital transformation of text (either printed or handwritten) of any language contained in images into machine editable text bypassing the conventional or traditional way of manually typing the text (Asif et al., 2014; Balaha et al., 2021a, 2021b, 2021c; Djaghbellou et al., 2022). OCR remains a prominent and active area of research in pattern recognition and computer vision (Ali et al., 2020; Djaghbellou et al., 2022; Eltay et al., 2020; Isheawy & Hasan, 2015). Additionally, automatic handwriting recognition is a sub-problem of OCR which has captured the interest of many researchers in the past decades due to many of its practical applications in day-to-day life ranging from bank cheques processing, document verification, data entry, to Natural Language Processing (NLP) tasks (Ahmed et al., 2020; Alrobah & Albahli, 2022; Ghadhban et al., 2020).

Handwritten text recognition is inherently more complex than printed text recognition. The primary and common challenge faced by handwritten text recognition models of different languages is rooted in the cursiveness and diversity of handwriting styles that significantly varies from one person to another (Patel & Thakkar, 2015; Sonkusare & Sahu, 2016; Yongping et al., 2022a). With the advent of new technologies, handwriting recognition has continued to evolve and expand over time. Despite significant progress by many

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researchers in the field over the years, unconstrained handwriting recognition remains an open challenge requiring further improvement (Elkhayati & Elkettani, 2021; Ghadhban et al., 2020; Porwal et al., 2013).

The development of handwritten text recognition has been a dynamic process, which has witnessed remarkable advancements across various languages among which are English, Chinese, and Hindi (Alrobah & Albahli, 2022; Geng et al., 2023; Shaiq et al., 2022; Srivastava et al., 2022; Yongping et al., 2022b). Such development, driven by advances in several machine learning and deep learning techniques together with data availability, has enabled improved character and word recognition, transforming tasks like digitizing historical manuscripts and aiding documents indexing (Odeh et al., 2022; Yee Fan et al., 2022).

In English, for example, neural network models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and transformer networks have been utilized, which has substantially enhanced the performance of cursive text (handwritten) recognition systems, which makes them proficient in converting handwritten English script into digital text with impressive precision (Perwej & Chaturvedi, 2012). On the other hand, Chinese handwriting recognition which faced the complex challenge of dealing with an extensive character set, and Hindi, with its unique Devanagari scripts, both of which have complex character structures and diverse writing styles, brought about the development of sophisticated frameworks that leverage both shape and context information, leading to significant breakthroughs in recognizing handwritten text in these languages making them more accessible via digital interfaces (Atul & Mishra, 2007; Geng et al., 2023; Jangid & Srivastava, 2018; Yongping et al., 2022a, 2022b).

Amidst this linguistic growth, and field of handwritten text recognition, Arabic presents its own set of unique challenges and vital importance. Arabic is not only a widely spoken language but also plays a substantial role in Islamic history and culture. Renowned for its complex calligraphy and several variations in styles of handwriting, Arabic script requires dedicated recognition models in order to capture its nuances and facilitate accessible education, seamless communication and preserve the essence of this rich linguistic heritage. This need is especially critical due to the widespread use of Arabic in administrative, cultural, and academic realms. Furthermore, given the significance of Arabic in religious texts, Islamic studies, and the wider Middle Eastern background, efficient HATR can facilitate efficient searching, indexing, and cross-referencing within vast collections of handwritten materials.

Hence, recognition of Arabic handwritten text is important for digitizing historical manuscripts, modern documents, and educational materials, contributing to the preservation of Arabic cultural heritage. As research in the recognition of other languages has shown, the ongoing development of advanced algorithms and models can undoubtedly pave the way for equally groundbreaking progress in the domain of Arabic handwritten text recognition.

Furthermore, the intricate morphology associated with Arabic writing system, characterized by a complex system of root letters, word patterns, and morphological changes, along with the extensive use of diacritics as well as rich calligraphic practice also poses significant challenges for accurate recognition (Djaghbellou et al., 2022; Kanoun et al., 2011; Lamghari & Raghay, 2021). Consequently, being among the world's most widely spoken languages, there is an increasing need for a reliable and robust HATR systems to ensure accessibility and enable effective communication for the Arabic-speaking population (Ahmed et al., 2021; Elkhayati & Elkettani, 2021; Haboubi et al., 2022).

Additionally, recognition of Arabic script has immense benefits as it is widely utilized in writing many other languages in some Muslim countries and societies such as Farsi, Urdu, Kurdish, Jawi, Pashto, Balochi, Sindhi, Uighur, and Kazakh among others. Even though the script remains consistent across these languages, specific modifications such as additional characters and diacritics are introduced to accurately represent the phonetic distinctions peculiar to each language (Porwal et al., 2013; Tanvir Parvez & Mahmoud, 2013).

Compared to printed Arabic text, handwritten Arabic text poses even greater challenges for recognition systems due to high variation in handwriting styles which differs from person to person (Albattah & Albahli, 2022). Great progress has been pursued in improving the recognition of Arabic handwritten text using several techniques with deep learning methods ranking at the forefront. However, existing models often struggle to handle complexities introduced by overlap between adjacent sub words and varying diacritics placements, resulting in errors and inaccurate interpretations.

This study seeks to review the recent trends and advances in the HATR realm. The contribution of the study includes the following:

1. We present a comprehensive review of the recent advances in the field of automatic Arabic handwriting processing, with a particular emphasis on offline handwritten text recognition, especially within the last 10 years.
2. We contextualize the broader field of handwritten Arabic text recognition by briefly introducing both online and offline paradigms, while primarily focusing our review on offline recognition techniques.
3. We highlight the key difficulties and challenges specific to offline Arabic text processing and recognition
4. Finally, we discuss the possible future directions in advancing offline HATR systems, including dataset

development, segmentation-free approaches, and contextual modeling.

The goal of the study is to offer a wide-ranging review of recent trends and advances in automatic Arabic handwriting processing especially within the last 10 years, and discuss available databases for research in the area. It reports many of the significant successes and promising approaches in the literature for offline recognition. The paper also highlights the various factors that make HATR difficult and challenging task, and finally, the future directions to be taken in the field.

2 Background

In this section, the background of Arabic language, its script description as well as the Arabic writing system is introduced.

2.1 Arabic Language

Language is considered the primary means of communication between humans. It is a structured method of conveying meaning both in written and spoken forms, that consists of both vocabulary and grammar. The Arabic script refers to the set of visual symbols or characters that are used in representing the spoken language in written form. However, the Arabic writing system (orthography) not only includes the script but also the grammar, punctuation, and other elements that make up the writing rules that governs the usage of those characters to represent the language. Thus, it is important for communication, conserving and sharing knowledge, history, and several forms of information recording over time (Pae & Wang, 2022; Plamondon & Srihari, 2000).

Dating back to first centuries of Islam, the Arabic script was exclusively used for inscribing Arabic language. However, with the passage of time and due to religio-cultural transfer, it eventually came to be used for writing other languages as earlier mentioned. The script's historical significance and widespread use across various regions have led to its adoption as the writing system for languages spoken by diverse communities, and is rated as the fifth top writing system used globally (Ali et al., 2020; Alotaibi et al., 2017; Balaha, Ali, Youssef, et al., 2021; Basha et al., 2022; Gader & Echi, 2022; Mumin & Versteegh, 2014; Porwal et al., 2013; Saidi et al., 2021).

Presently, Arabic is the official language of many countries across the Middle East and North Africa (Ali et al., 2020; El Atillah et al., 2022; Porwal et al., 2013). It is among the languages spoken widely across the globe with over 500 million speakers, both native and non-native speakers (Balaha, Ali, Youssef, et al., 2021; Ghadhban et al., 2020; Saidi et al., 2021).

2.2 Arabic Writing System Description

The Arabic script writing format, as shown in Fig. 1, is from right to left, unlike English and other scripts (Ahmed et al., 2020). The script consists of twenty-eight (28) characters (alphabets) which are all consonants with no upper- or lower-case representation (Ahmed et al., 2020; Ali et al., 2020; Alrobah & Albahli, 2022; Balaha et al., 2021a, 2021b, 2021c; Balaha, Ali, Youssef, et al., 2021; Lorigo & Govindaraju, 2006; Porwal et al., 2013; Saidi et al., 2021). Arabic characters exhibit two to four contextual shapes depending on their position in a word, (whether at the start, middle, end of a word, or as an isolated letter) (refer Table 1) thereby increasing the possible classifications from 28 to as many as 100 (Ali et al., 2020; Alrobah & Albahli, 2022; Balaha, Ali, Youssef, et al., 2021; Kanoun et al., 2011). Additionally, dots are inherent part of Arabic characters.

The Arabic script has sixteen (16) main strokes. While identical strokes are used for different alphabets, the number of dots associated with the alphabet distinguishes each alphabet from another (Ahmed et al., 2020; Balaha et al., 2021a, 2021b, 2021c). Also, a unique character called Hamzah (ء), can function either as standalone character or sub component located under, inside, or directly above some primary characters (Alrobah & Albahli, 2022; Balaha, Ali, Youssef, et al., 2021).

Arabic script in printed formats as well as handwritten, appears cursive with the characters connected horizontally in a word by an imaginary line known as "the baseline" as demonstrated in Fig. 2. Character parts that extend above the baseline are referred to as "ascenders" while those that extend below the baseline are known as "descenders" (Balaha et al., 2021a, 2021b, 2021c; Lorigo & Govindaraju, 2006). Arabic words are composed of multiple sub-words (refer Fig. 3) which are also referred to as pieces of Arabic Words (PAWs) (Balaha, Ali, Youssef, et al., 2021).

Arabic script features intricate connections between characters, forming ligatures that can vary greatly across different styles of handwriting. Ligature (also known as "ta'liq") are very common in Arabic writing. It is the blend of two or more characters into a single character; a typical example is the "laam-alif" (لا) combination (see Fig. 4).

Unlike Latin script, which uses vowel characters directly within words, Arabic relies heavily on diacritics

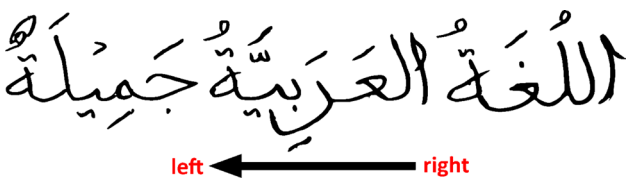


Fig.1 Direction of Arabic text

Table 1 Contextual forms of Arabic characters

Character name	Isolated	Start	Middle	End
Alif	ا	ا	ـا	ـا
Baa	ب	بـ	ـبـ	ـب
Taa	ت	تـ	ـتـ	ـت
Thaa	ث	ثـ	ـثـ	ـث
Jeem	ج	جـ	ـجـ	ـج
Ha	ح	حـ	ـحـ	ـح
Khaa	خ	خـ	ـخـ	ـخ
Daal	د	د	ـد	ـد
Dhaal	ذ	ذ	ـذ	ـذ
Raa	ر	ر	ـر	ـر
Zay	ز	ز	ـز	ـز
Seen	س	سـ	ـسـ	ـس
Sheen	ش	شـ	ـشـ	ـش
Saad	ص	صـ	ـصـ	ـص
Daad	ض	ضـ	ـضـ	ـض
TTaa	ط	طـ	ـطـ	ـط
Zaa	ظ	ظـ	ـظـ	ـظ
Ayn	ع	عـ	ـعـ	ـع
Ghayn	غ	غـ	ـغـ	ـغ
Faa	ف	فـ	ـفـ	ـف
Qaaf	ق	قـ	ـقـ	ـق
Kaaf	ك	كـ	ـكـ	ـك
Laam	ل	لـ	ـلـ	ـل
Meem	م	مـ	ـمـ	ـم
Noon	ن	نـ	ـنـ	ـن
Haa	ه	هـ	ـهـ	ـه
Waw	و	و	ـو	ـو
Yaa	ي	يـ	ـيـ	ـي

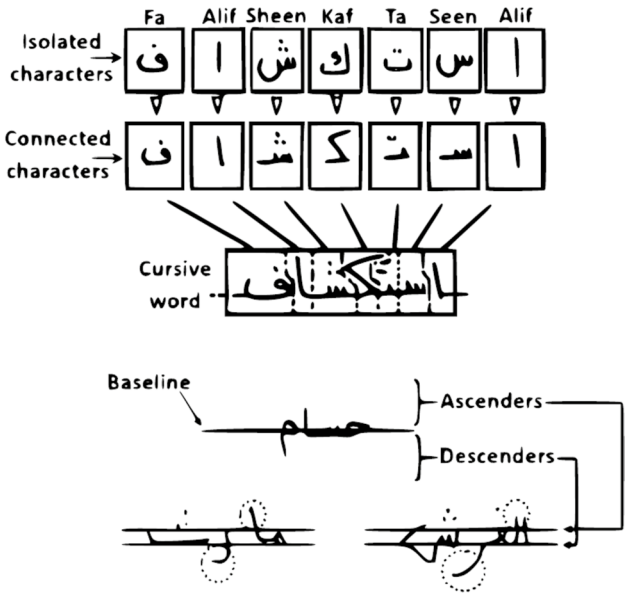


Fig. 2 Cursive nature of Arabic characters

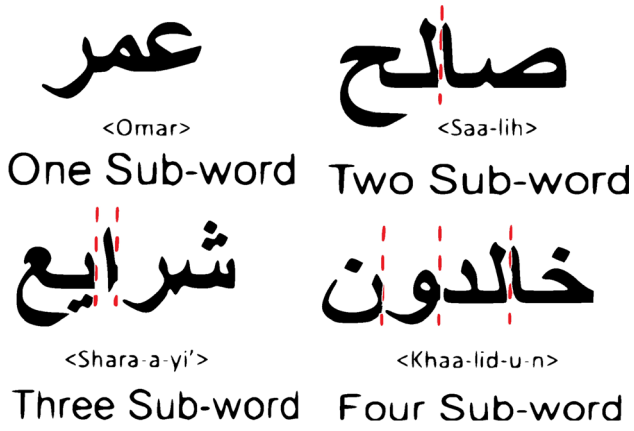


Fig. 3 Sub-words in Arabic text also known as PAWs

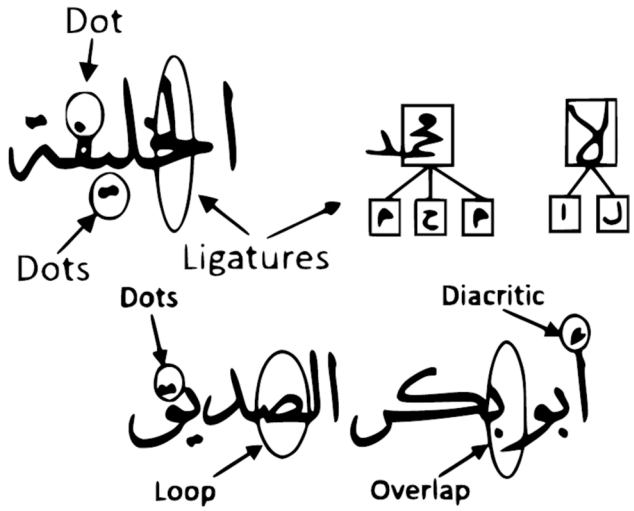


Fig. 4 Characteristics of Arabic script

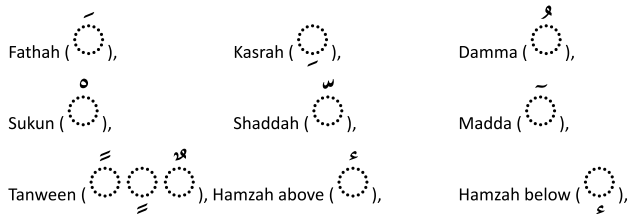


Fig. 5 Diacritics (Tashkeel) in Arabic script

to express short vowels, pronunciation cues, and certain grammatical features. Arabic diacritics are small marks placed above or below characters to indicate pronunciation, short vowels, and grammatical features (see Fig. 5). While they enhance readability, missing or misplacing them can drastically change a word's meaning, posing a major challenge for recognition systems. (Alrobah & Albahli, 2022; Lamghari & Raghay, 2021; Mohd et al.,

2021; Nahla Ibrahim Youssef, 2022; Tanvir Parvez & Mahmoud, 2013).

3 Arabic Text Recognition

In this section, an overview of Arabic text recognition systems is provided, including categories in addition to the general steps involved in the recognition process.

3.1 Recognition Categories

Handwriting recognition is generally classified into two; namely online and offline recognition (Fasha et al., 2020). Similarly, Arabic handwriting Recognition, as shown in Fig. 6, is classified into online and offline recognition. While online recognition is concerned with the available trajectory data, offline recognition is concerned with data from images of scanned documents (Plamondon & Srihari, 2000). Furthermore, the handwriting recognition process can function in a manner that is either tailored to specific writers ('writer-dependent') or generalized to multiple writers ('writer-independent') (Alwajih et al., 2022; Djaghbellou et al., 2022; Sepahvand et al., 2017). Although both recognition paradigms are explored in literature, the present review focuses on offline recognition. The following subsection briefly discusses online HATR to contextualize their differences.

3.1.1 Online Recognition

The online handwritten text recognition (HATR) involves recognition of Arabic characters as they are written in real time using a digital pen or fingertip on touchscreen devices such as a computer, mobile phone, tablets, Personal Digital Assistants (PDA) or digital writing pads, simply by capturing the temporal sequence of points that are used to analyze the path of the author's handwriting and then extract features such as the direction of the stroke, the length and the curvature. Subsequently, the word or character inscribed by the author is recognized using these features (Lorigo & Govindaraju, 2006; Nahla Ibrahim Youssef, 2022). Several approaches have been proposed and experimented in the literature (Alwajih et al., 2022; Hamdi et al., 2023; Othmen et al., 2022; Sepahvand et al., 2017; Tagougui, Boubaker, et al., 2013; Tagougui et al., 2014) for different challenges concerning online HATR, likewise, many works have been summarized in some review papers (Tagougui, Kherallah, et al., 2013a, 2013b).

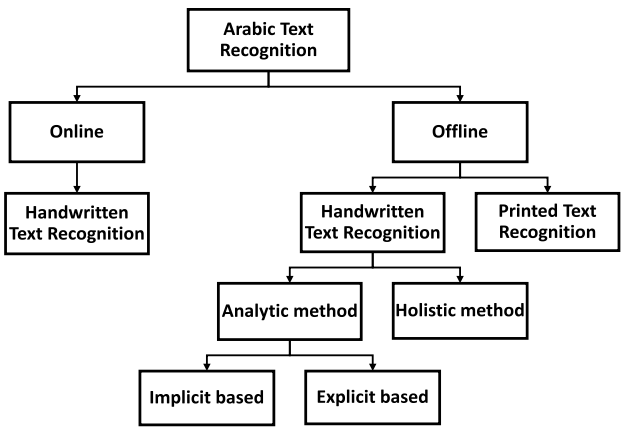


Fig. 6 Categorization of Arabic text recognition

3.1.2 Offline Recognition

Unlike online recognition, in the offline recognition, there is no direct interaction with the writer. The offline HATR involves recognizing Arabic text from images of existing text of both printed and handwritten documents in which the only available information is the pixel values (Fasha et al., 2020; Lorigo & Govindaraju, 2006; Othmen et al., 2022). Usually, the online recognition is easier as there are more information available for the recognition process (Lorigo & Govindaraju, 2006; Nahla Ibrahim Youssef, 2022). Therefore, due to the differences in the available information as well as the processing, the two modes (online and offline) require different approaches for analysis and recognition, although similar techniques are being used.

3.2 General Stages of Arabic Text Recognition System

Typically, HATR involves five different stages: Preprocessing stage, Segmentation stage, Feature extraction stage, Classification stage, then Recognition/post-processing stage. Understanding these recognition stages is fundamental to developing effective recognition models and consequently advancing the HATR domain. These stages are shown in Fig. 7. However, image acquisition is generally required to gather and convert the hardcopy text documents into digital images. Moreover, preprocessing is required for enhancing the acquired digital images' quality for a smooth and easy segmentation and features extraction. The digital images may be acquired through camera captures or scanning, as a result, it is normally affected by noise, quality degradation, skewness, etc.

3.2.1 Preprocessing Stage

Preprocessing the acquired text images is an essential pre recognition task as the image quality significantly affects the recognition accuracy (Ahmed et al., 2012; Al Abodi & Li, 2014). This involves applying various operations on the

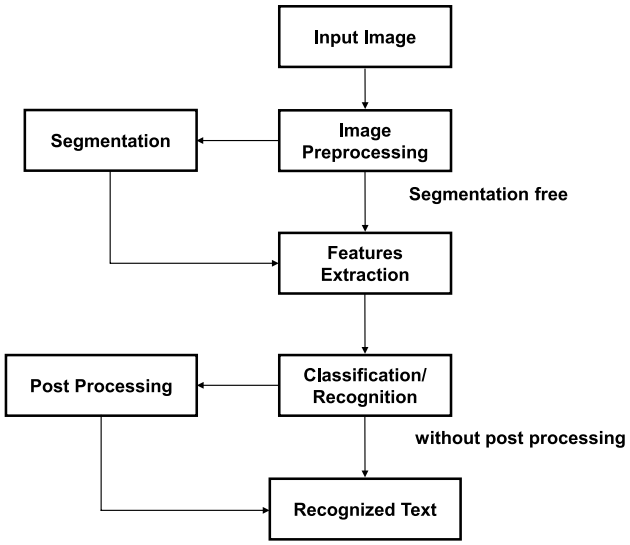


Fig. 7 Stages of Arabic text recognition

raw images (based on the image quality level) to refine as well as to ensure that the input data is cleaner and to convert the images into more compact representation suitable for the recognition process, such as skeletonization or thinning (reducing the character shape to a width of one pixel in order to show centerlines of the character), binarization (to convert an image from grayscale colored to binary based on a certain threshold), normalization (resizing character to a consistent height and width and ensure it is centered), smoothing (filtering to reduce noise and irregularities in the image), baseline detection (identifying the baseline; line on which most characters align) (Ahmed et al., 2021; Djaghbellou et al., 2022; Lorigo & Govindaraju, 2006).

3.2.2 Segmentation Stage

Following preprocessing in the recognition pipeline is segmentation, which involves segmenting text image into small elements or components i.e., separating a paragraph into text lines or words within a text page (external segmentation), or separating words into sub words or individual characters (internal segmentation) (Ahmed et al., 2021; Arica & Yarman-Vural, 2001). Segmentation is among the key difficulties in HATR, which is also reported to be the main contributor of increase in recognition error rate (Ahmed et al., 2021; Alghamdi & Teahan, 2018; Khorsheed, 2002).

Very often, there exist cases of character overlaps and ligatures in handwritten Arabic text due to its cursive and connectiveness nature, which can lead to erroneous segmentation outcome such as under or over segmentation, which then leads to misclassification of characters which subsequently affects the recognition process (Alginahi, 2013).

Segmentation techniques such as contour detection, connected component analysis, projection profile or histogram, artificial neural networks, morphological operations, and template matching are usually applied to the preprocessed data to identify and separate words or characters. For text line to words segmentation, the algorithm may look for spaces (long spaces between words), punctuation marks, or other cue to determine the words boundaries, and may detect and identify segmentation points (character ending and the next character beginning) to segment words into individual characters (Balaha et al., 2021a, 2021b, 2021c; Djaghbellou et al., 2022; Najam & Faizullah, 2023; Plamondon & Srihari, 2000; Zoizou et al., 2020).

3.2.3 Features Extraction

The extraction of features is a crucial and most essential stage in recognizing handwritten text (Balaha et al., 2021a, 2021b, 2021c; Djaghbellou et al., 2022; Faizullah et al., 2023). Features are distinctive patterns, attributes or characteristics that capture essential information from the input image data and transform them into a vector of features that enables the recognition system to accurately discriminate one character from another (Ahmed et al., 2021). In this stage, the objective is capturing unique characteristics or features of the Arabic letters that differentiate between one character from another to aid in correct classification of the words or characters.

The various types of features are generally characterized into statistical features or points distribution, structural features, gradient based features, and feature space transformations (refer Table 2).

The statistical features technique is based on extracting features by analyzing the statistical distribution of pixel values within a character. These features include zoning, moments, n-tuples (diacritics and their positions), crossing and branching points, loops, characteristics loci and segments, and typically tolerate distortions and variations in style with high speed and low complexity (Ahmed et al., 2021; Faizullah et al., 2023).

The structural features primarily capture/extract features describing the topological and geometrical attributes within characters i.e., features depicting the physical structure of the character. These features such as strokes, loops, branching points, endpoints, the presence and count dots along with their respective positions, are especially useful for scripts with complex strokes and diacritics like the Arabic script. As discussed earlier, several characters in Arabic script share similar base shapes and differ only by dots, making structural feature extraction essential (Ahmed et al., 2021; Djaghbellou et al., 2022; Lorigo & Govindaraju, 2006).

Techniques using gradient based features capture details regarding changes in pixel intensity within a character or

Table 2 Feature extraction in handwritten Arabic text recognition (HATR)

Type	Description	Example(s)	Advantage
Statistical	Analyzes statistical properties such as points distribution of pixel values within a character	Zoning, moments, n-tuples crossing and branching points, loops, characteristics loci and segments	Typically tolerate distortions and variations in style with high speed and low complexity
Structural	Topological and geometrical features (physical structure of the character)	Strokes, loops, branch points, number of dots and their positions, end points or other	Useful for Arabic script complex strokes and diacritics as well as characters with similar base shapes
Gradient based	Details about changes in pixel intensity within a character or locality	Highlights character edges and contour	Useful for discriminating features
Feature Space Transformation	Reduces dimensionality of resulting feature vectors	Fourier transform, Hough transform, Gabor transform, PCA, DCT or other	Mitigates high dimensionality and hence the problem of overfitting

locality. Mainly, these features are effective in highlighting the character edges and contour, which can be useful discriminating features. A common gradient based feature is the Sobel operator which reveals edge information by computing the gradients in the horizontal and vertical directions (Chang et al., 2023; Jacobs, 2005; Jing et al., 2022; Sahlol et al., 2020; Chen & Zhao, 2023; Venugoban & Ramanan, 2014).

Feature extraction often yields feature vectors of high dimensionality which can incur computational costs and increase the likelihood of overfitting. Hence, to mitigate these issues, techniques for reducing dimensionality, including Principal Component Analysis (PCA) or Linear Discriminant Analysis (LDA), Fourier and Hough Transform, Discrete Cosine Transform (DCT), Walsh Hadamard, Karhunen Loeve, Wavelets, and Gabor Transform among others may be used in reducing the feature space and at the same preserve the most discriminating data (Ahmed et al., 2021; Balaha et al., 2021a, 2021b, 2021c; Lawgali, 2015).

However, features extraction comes with its own challenges; choice of features (leaving out irrelevant attributes), robustness to variation (noise, writing style, size, orientation, etc.), and computational efficiency (especially for real time or resource constrained applications). Hence, the features extraction method can be assessed based on its sensitivity to noise and distortion as well as ease of use and implementation (Sahlol et al., 2020).

Furthermore, selecting appropriate features and effective features extraction method has huge impact on achieving high recognition rate and computational efficiency (Lawgali, 2015). Fortunately, advances in deep learning methods like CNNs have enabled automatic feature learning and by this means, the need for manual or handcrafted feature engineering has reduced in many recognition models.

3.2.4 Classification Stage

This stage identifies individual character or word and assigns it to the right character or word class. During classification, the system determines the most likely class that matches the input based on extracted features. This decision-making process is guided by machine learning algorithms that are appropriately selected for the recognition. Two processes are normally implemented in this stage: the first is the learning process (training); in which the model being selected for the classification is trained with the extracted vector features from the training dataset and then through learning, the model recognizes the patterns and understand the relationships between the learned features and each label that corresponds to it (supervised learning) (other types of learning include unsupervised and reinforcement learning), which typically involves using optimization techniques that adjusts the parameters of the model and minimizes classification errors. Next process is testing; in which evaluation of the model's performance is carried out (on the test data) by assessing how reliably it adapts to unfamiliar or untrained data. Several classifiers have been utilized for the task of recognizing Arabic handwriting including Hidden Markov Models (HMMs), Artificial Neural Networks (ANNs), support vector machines (SVMs), k-Nearest Neighbor (kNNs), and deep learning models e.g., CNNs and RNNs (Ahamed et al., 2020; Ahmed et al., 2021; Djaghbellou et al., 2022; Lawgali et al., 2014).

3.2.5 Post Processing Stage

In some cases, for the purpose of enhancing the accuracy of recognition, it may be necessary to employ post processing techniques to refine and validate the classification outcomes. Basically, the primary purpose of the post processing is error correction which involves identifying and correcting recognition errors i.e., misclassified words or characters. Some common techniques for post processing include error correction methods using e.g., dictionary or lexicon based; such that words not found in the dictionary are flagged as potential errors, Damerau-Lavenshtein distance, and statistical language models such as n-grams and RNNs, and contextual

post processing such as the use Conditional Random Fields (CRFs) for label smoothing and enforcing global consistency across predicted sequence. For Arabic text recognition, this could involve leveraging Arabic linguistic knowledge which can span various levels from character level to higher semantic level (Ahmed et al., 2021; Lawgali et al., 2014; Nahla Ibrahim Youssef, 2022). Successful post processing significantly improves the overall accuracy and usability of HATR systems.

3.3 Methodological Approach of Recognition

Arabic text recognition is traditionally based on two distinct methods that is the analytic and holistic methods.

In the analytic method of recognition (segmentation based), the handwritten text is separated into individual components such as characters or sub words to isolate and separately analyze the individual components within the text (Alghamdi & Teahan, 2018; Arica & Yarman-Vural, 2001). Analytic methods rely on segmenting words into characters or sub-words before recognition. This can be done explicitly; by isolating components for individual analysis, or implicitly; where segmentation and recognition occur simultaneously (Alghamdi & Teahan, 2018; Eltay et al., 2020).

Dating back to the 1970s and 1980s when investigations in the area of Arabic character recognition initially began, the segmentation phase was basically the most crucial part of the recognition process. Some of the early works on Arabic word segmentation is a structural recognition method proposed by (Al-Emami & Usher, 1990; Almuallim & Yamaguchi, 1987; Amin & Mari, 1989; Goraine et al., 1992).

In contrast, holistic methods treat the entire word or subword as a single unit, bypassing character-level segmentation. (Alghamdi, 2019). These approaches reduce segmentation errors and is suitable for capturing context within the text. Some studies have reported improved recognition rates using holistic models. However, this approach may only be suitable for small size lexicon (limited vocabulary) (Eltay et al., 2020).

The difference between the analytic and holistic methods is basically the extent of the segmentation applied. While the analytic methods employ segmentation into individual characters, the holistic methods typically only segment text line into words (or sub words in case of Arabic text) and subsequently, the classifier such as HMM, ANN, RNN or Connectionist Temporal Classification (CTC), identifies the whole word. Therefore, distinctive features of the word or sub word are extracted (Al-Sheikh et al., 2020).

3.4 Review of Recognition Methods

Handwritten Arabic Text Recognition (HATR) has evolved significantly over the past decade, transitioning from handcrafted techniques and traditional machine learning models to more sophisticated deep learning architectures and hybrid approaches. This section presents a categorized review of recognition methods used in HATR: traditional, deep learning, and hybrid techniques. Each category is examined in terms of its methodology, datasets employed, achievements, and inherent limitations, which are summarized in Table 3. A comparative analysis concludes the section, highlighting the strengths and limitations of reviewed methods.

3.4.1 Traditional Methods

Traditional recognition methods primarily involve handcrafted feature extraction combined with classical machine learning classifiers. Dominant before the widespread adoption of deep learning, these methods typically follow a multistage pipeline: preprocessing, segmentation, feature extraction, and classification.

Handcrafted features are manually designed descriptors such as zoning, projection profiles, chain codes, stroke direction, and loop or junction detection (see Table 2). These features aim to capture structural properties of Arabic script, including character connectivity, baseline positioning, and stroke curvature. Due to the complexity and variability of Arabic handwriting, feature design required significant domain expertise. The extracted features were often passed to basic classifiers such as rule-based engines or statistical models, which limited their robustness and generalization across diverse handwriting styles (Khedher et al., 2005).

The study of (Cheung et al., 2001) proposed a segmentation algorithm capable of handling horizontally overlapped sub-words. By leveraging structural cues such as Characteristic Defining Points (CDPs) and connectivity features, the method avoids explicit segmentation points, achieving 90% accuracy at 20 characters per second. However, the system was challenged by deformed characters and visual similarity among letters, which the authors mitigated through a user-interactive feedback loop.

Similarly, (Lorigo & Govindaraju, 2005) presented a rule-based over-segmentation method refined by shape heuristics. Techniques like the gradient method and "down-up" stroke detection were used to detect accurate segmentation points, achieving 92.3% accuracy on a 200-image dataset.

In (Mozaffari et al., 2005), they combined structural and statistical features for digit recognition. Skeleton-based primitives were encoded and aggregated into global descriptors, statistically characterized by changes in X and Y coordinates. PCA was applied to normalize feature lengths, improving classification consistency.

The study of (Al-shatnawi & Omar, 2014) reviewed thinning (skeletonization) as a key preprocessing step in Arabic text recognition. They highlighted its role in baseline

Table 3 Summary of Related Works in HATR

Study	Method	Dataset	Recognition Level	Accuracy	Limitation	Advantage
Cheung et al. (2001)	Rule-based segmentation using CDPs	Not specified	Sub-word/Word	90%	Misrecognition of similar or deformed characters	CDP-based horizontal segmentation
Alma'adeed et al. (2004)	Global feature rule-based engine + HMM	Not specified	Word	Not stated	Early rule-based stage may reduce adaptability	Initial fusion of structural and statistical models
Lorigo and Govindaraju (2005)	Rule-based segmentation with shape info	200 images (custom)	Character segmentation	92.3% (segmentation points)	Focused only on segmentation	Loop/stroke-based segmentation
Mozaffari et al. (2005)	Statistical + Structural features + PCA	Not specified	Arabic digits	Not specified	Handcrafted feature dependency	Statistical characterization of shapes
Al-Shatnawi and Omar (2014)	Thinning/skeletonization review	Not applicable	Character (Preprocessing)	Not applicable	Connectivity issues, noise sensitivity	Preprocessing insights
Al Abodi and Li (2014)	Geometric feature extraction	Not specified	Character	Not specified	Simplistic vector encoding	Spatial stroke vector encoding
Elleuch et al. (2016)	CNN + SVM, with dropout + elastic deformation	HACDB, IFN/ENIT	Character	5.83% (HACDB), 7.05% (IFN/ENIT)	No sequence-level modeling	CNN-SVM integration with strong generalization
Ali and Suresha (2017)	Skew correction + Morph filtering	Not specified	Preprocessing	Improved visual quality	Limited to noise correction	Handles Gaussian/Impulse noise
El-Sawy et al. (2017)	CNN (basic architecture)	AHCD (16,800 samples)	Character	94.9%	No sequence modeling, isolated characters only	Simple, reproducible CNN + dataset release
Maalej and Kherallah (2018)	CNN + BLSTM + CTC	IFN/ENIT (extended)	Word (Seg-free)	92.21%	Limited vocabulary	Handles cursive and ligatures in end-to-end setup
Doush et al. (2018)	Post-processing model: error model + context model	Custom (9000 lines)	Line/post-processing	Not stated	Focused only on post-processing	AOCR post-correction model
Ali and Suresha (2019a)	Vertical segmentation via stroke geometry	Not specified	Character	Not specified	Limited generalizability	Robust to touching characters
Sheikh et al. (2020)	Region-based diacritic segmentation	Not specified	Diacritics	Not quantified	Diacritic size variation	Region-based analysis of diacritics
Altwaijry and Al-Turaiki (2020)	CNN with Hijja dataset	AHCD, Hijja (children's writing)	Character	97% (AHCD), 88% (Hijja)	Poorer performance on children's writing	Hijja dataset reveals age-based variability
Sobhi et al. (2020)	CNN–RNN encoder–decoder + attention	KAFD	Word	89.82%	Limited parallelism due to RNNs	Font-robust model with attention
Khosrobeygi et al. (2020)	Rule-based post-processing	Persian OCR (300 images)	Line-level	93% (Khana), 91% (Bina)	Language-specific model	Post-processing boosts accuracy
Balaha et al., (2021a, 2021b, 2021c)	Modular CNNs (HMB1, HMB2)	HMBD, CMATER, AIA9K	Character (contextual)	98.4% (HMB1 + augmentation)	No sequence or attention modules	Large-scale training with contextual forms
Balaha et al., (2021a, 2021b, 2021c)	Deep CNN + Transfer Learning + Genetic Algorithms (HMB-AHCR-DLGA)	HMBD (54K), AIA9K, others	Character	92.88%	No sequence modeling	Optimized pipeline with pretrained model evaluation

Table 3 (continued)

Study	Method	Dataset	Recognition Level	Accuracy	Limitation	Advantage
Rahal et al. (2021)	Deep Sparse Autoencoder + Bag-of-Features + HMM	P-KHATT, APTI, IFN/ENIT, MNIST	Character/Sequence	Not specified	No RNN/Transformer-based modeling	Adaptable to resolution variance
Saidi et al. (2021)	Structural features + RBF neural network	Not specified	Character	Not stated	Unclear dataset and evaluation method	Generalization using RBF
Bouchakour et al. (2021)	CNN + histogram + Gabor + contour segmentation	Not stated (printed text focus)	Character (Printed)	Not stated	Printed Arabic focus	Effective feature fusion
Alrobah and Albahli (2021)	CNN (HMB1/HMB2) + SVM/XGBoost	HIJJA	Character	96.3%	Limited to isolated characters	Efficient SVM integration with robust generalization
Aliwy and Al-Sadawi (2021)	Statistical language model-based error correction	Not stated	Post-processing	Up to 98% OCR accuracy gain	No recognition architecture integrated	Language model enhances post-correction
Mostafa et al. (2021)	CNN (ResNet101) + Transformer + Mask R-CNN	Custom 30M lines (subset trained)	Line	CER: 7.27%, WER: 8.29%	Limited training set (15K lines); heavy computation	Layout- and diacritic-aware recognition
Mostafa et al. (2022)	BEiT ViT + Transformer Decoder + Detectron-2	270M words, 13 fonts/styles	Line	CER: 4.46%	High resource demands	Fully end-to-end, high generalization
Eltay et al. (2022)	GAN-based data augmentation	IFN/ENIT, AHDB	Word	Improved over baseline	GAN realism not benchmarked	Targeted augmentation for rare classes
Ali and Mallaiah (2022)	CNN + SVM + M3CE Loss + SPPnet + NIN modules	AHDB, AHCD, HACDB, IFN/ENIT	Character/Word	Up to 99%, 98.58% (IFN/ENIT)	No sequence/linguistic modeling	Enhanced classifier with spatial invariance
Lamtougui et al. (2022)	CNN + BLSTM + CTC	Not specified	Word (Seg-free)	Not stated explicitly	Dataset unspecified; limited detail	Segmentation-free
Gader and Echi (2022)	CNN + BLSTM + Attention + CTC	IFN/ENIT, IAM	Word (Seg-free)	97.1%, CER 2.9%	Limited to curated datasets	Interpretable, language-independent generalization
Nayef et al. (2022)	CNN + OLReLU activation	AHCD, AIA9K, HIJJA, MNIST, custom	Character	Up to 99% (AHCD)	Isolated characters only	Activation tuning; generalizable architecture
Ullah and Jamjoom (2022)	Shallow CNN with dropout & BN	AHCD	Character	96.78%	No contextual modeling	Efficient and reproducible architecture
Abdo et al. (2022)	Horizontal projection + thinning	Not specified	Character	Not specified	Traditional segmentation method	Multi-stage segmentation pipeline
Momeni and BabaAli (2022)	Transformer, Transformer Transducer (RNN-T)	KHATT	Word	Outperformed RNN + CTC	No real-world noise considered	Recurrence-free sequence modeling with attention
Mehdi and Abdelghani (2022)	Capsule Network with residual CNN input	IFHCDB, AHCD, AIA9K	Character	Up to 97.57%	High computation cost	Robust to diacritics and stroke variations
Maghraby and Samkari (2023)	CNN + GRU + CTC	Custom (110 diacritic-rich word images)	Word with diacritics	32.5% CRR, 0% WRR	Small dataset size	Diacritic-aware recognition

Table 3 (continued)

Study	Method	Dataset	Recognition Level	Accuracy	Limitation	Advantage
AlShehri (2024)	Deep CNN (DeepAHR)	AHCD, HIJJA	Character	98.66% (AHCD), 88.24% (Hijja)	No sequence modeling	Strong benchmark across handwriting styles
Aabed and Khairaldin (2024)	DCNN + BLSTM + CTC (Segmentation-Free)	KHATT	Line	84% (char), 71% (word)	No attention or transformers	Practical design for real-world applications

detection, segmentation, and feature extraction. Challenges included maintaining stroke connectivity, suppressing spurious tails, and handling noise during skeleton generation.

A geometric approach by (Al Abodi & Li, 2014) simplified characters to single-pixel-width strokes and encoded them as vertical and horizontal vectors. This method aimed to standardize diverse handwriting styles into geometric representations suitable for classification. Ali and Suresha (2017) proposed a skew correction technique using binary morphological operations, enhancing recognition by addressing Gaussian and impulse noise. Later, they introduced a vertical segmentation method (Ali & Suresha, 2019a) based on stroke width and character geometry to better manage ligatures in Arabic script. In (Sheikh et al., 2020), a region-based method for extracting diacritics, using connected components and regional analysis is proposed. Although effective, the method faced challenges in determining size thresholds due to diacritic variability.

Post-processing has also been explored in (Khosrobeygi et al., 2020). They developed a feature-rule-based filter to refine text-line outputs, improving recognition accuracy on complex Persian scripts. Similarly, (Abdo et al., 2022) applied thinning and geometric analysis for segmentation, while (Qaroush et al., 2022) used a hierarchical projection strategy to extract text lines.

Recognizing the limitations of basic classifiers, researchers began incorporating more sophisticated algorithms while retaining manual feature extraction. This marked a shift toward the second wave of traditional methods.

Classical machine learning classifiers such as k-Nearest Neighbors (k-NN), Support Vector Machines (SVM), and Hidden Markov Models (HMM) became prominent:

- **K-NN** is a simple, instance-based classifier used with handcrafted features like projection histograms or zoning. While effective in small feature spaces, it is sensitive to noise and computationally intensive on large datasets. Nonetheless, it serves as a strong baseline and is often included in hybrid systems.
- **SVMs** gained popularity for their ability to handle high-dimensional, non-linear data. When paired with features such as Gabor filters or intensity histograms, they offer strong generalization, such as the work in (Elleuch et al., 2017), demonstrating notable gains in character classification. However, optimal performance requires careful kernel tuning and quality feature input.
- **HMMs** have been extensively used in sequence modeling for Arabic script. By converting 2D images into 1D observation sequences, HMMs model the temporal nature of handwriting. Studies like (Ahmad & Fink, 2016; Dehghan et al., 2001; Eraqi & Abdelazeem, 2012; Khorsheed, 2003, 2007) demonstrated HMMs' strength in capturing contextual variation and character

transitions. Khorsheed's work introduced a segmentation-free HMM model using ergodic character states for entire word recognition. However, the need for handcrafted features and large annotated datasets limits their scalability.

In (Dehghan et al., 2001), a holistic HMM system using chain code histograms and self-organizing feature maps is proposed, achieving 65% accuracy on a dataset of 198 Iranian city names. Similarly, (Khorsheed, 2003) developed an HMM with 296 states across 32 character models, capable of recognizing entire words without segmentation.

In (Tamen et al., 2017), the authors introduced a multiple classifier system using Chebyshev Moments and Statistical Contour Features (SCF) from IFN/ENIT. Classifiers such as SVM, multilayer perceptron (MLP), and extreme learning machine (ELM) were fused with strategies like Dempster-Shafer and Borda Count, achieving 96.82% accuracy. Despite lacking deep learning integration, the ensemble approach proved robust for complex scripts.

In (Ali & Suresha, 2019b), they proposed a layered fusion framework using four feature types (e.g., run-length matrix, moment invariants) and three classifiers (SVM, modified quadratic discriminant function, random forest), achieving over 99% accuracy on AHCD and HACDB. Although limited to character-level tasks, the framework handles overlapping and ligature-heavy characters effectively.

In (Zoizou et al., 2022), they introduced the MOJ-DB dataset with a contour-based segmentation algorithm for historical Arabic sub-words. Their preprocessing and segmentation techniques improved extraction accuracy and support future deep learning work, though the study focuses primarily on handcrafted features.

3.4.2 Deep Learning Methods

Since its introduction, deep learning has profoundly transformed HATR, shifting the field away from handcrafted features toward automated, data-driven architectures. Early successes with CNNs led to widespread adoption, particularly for isolated character recognition. Over time, the need to capture sequential and contextual dependencies gave rise to more advanced models such as RNNs, attention-based mechanisms, and Transformers. These deep learning techniques have enabled end-to-end training, segmentation-free recognition, and significantly improved generalization across varied scripts and datasets. This section reviews the major deep learning architectures applied in HATR, tracing their evolution while highlighting their contributions, strengths, and limitations.

- Artificial Neural Networks (ANNs)

ANNs marked the shift from rule-based to data-driven modeling in early HATR. These networks were particularly effective for isolated character classification, especially when combined with handcrafted features, as seen in the works of Hamid and Haraty (2001) and Haraty and Ghaddar (2003). Their simplicity and adaptability to small datasets were beneficial in low-resource settings. However, ANNs lack spatial awareness and are inadequate for modeling sequential dependencies, limiting their applicability to word- and line-level recognition tasks.

- Convolutional Neural Networks (CNNs)

CNNs have become the backbone of HATR due to their ability to learn spatial hierarchies directly from raw images. Their layered structure captures both local and global visual patterns, making them highly effective for character- and word-level recognition. CNNs are robust to handwriting variability and noise and have consistently outperformed earlier methods in accuracy. However, they inherently limited in modeling sequential data, necessitating integration with temporal models for contextual understanding. Notably, pre-trained CNNs like VGGNet and ResNet have been leveraged through transfer learning to improve performance on limited datasets.

Transfer learning enables the adaptation of large, pre-trained models (e.g., ResNet50, VGG19) to specific Arabic handwriting tasks. This approach accelerates training and improves generalization, particularly useful when labeled data is scarce.

- Recurrent Neural Networks (RNNs) and Variants

RNNs, especially Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), are widely adopted for modeling the sequential nature of Arabic scripts. These architectures process sequences step-by-step, capturing temporal dependencies crucial for word- and line-level recognition. LSTMs are effective for long-range context, while GRUs offer computational efficiency. Nonetheless, RNNs suffer from slow training and vanishing gradients and are less scalable compared to transformer models. Their integration with CNNs laid the foundation for hybrid models capable of end-to-end recognition.

- Connectionist Temporal Classification (CTC)

CTC loss has enabled alignment-free sequence modeling in HATR, eliminating the need for character segmentation. By allowing the network to learn output-label alignments implicitly, CTC facilitates training of models like CNN-RNN hybrids. While it enhances flexibility and simplifies pipeline design, CTC decoding can be

error-prone in noisy predictions and often benefits from language model support.

- Capsule Networks (CapsNets)

Capsule Networks were introduced to overcome CNN limitations in modeling spatial relationships and hierarchical structures. In Arabic handwriting, they offer better discrimination for visually similar letters (e.g., س and ش) and characters with diacritics. CapsNets preserve orientation and relative positioning of features through dynamic routing, making them suitable for scripts with complex structures. However, they are computationally intensive and remain less widely adopted in large-scale HATR systems.

- Transformer-Based Models

Transformers have gained traction in HATR for their ability to model long-range dependencies via self-attention mechanisms. Unlike RNNs, transformers process input sequences in parallel, improving training efficiency and contextual understanding. They are particularly effective in handling long contextual dependencies present in cursive Arabic lines. Vision Transformers (ViTs) are increasingly replacing CNNs in encoder roles. However, these models require extensive training data and computational resources, posing challenges for low-resource scenarios.

- Attention Mechanisms and Sequence-to-Sequence Models

Attention mechanisms, integrated into sequence-to-sequence (Seq2Seq) architectures, enhance recognition by dynamically focusing on relevant image regions. This is particularly useful for cursive Arabic, where character position and shape vary widely. Attention modules are now commonly incorporated into RNNs, CNNs, and Transformers, improving both accuracy and interpretability. Despite their advantages, attention-based models are computationally demanding and require careful design and tuning.

In (El-Sawy et al., 2017), the authors introduced a simple yet influential CNN model for isolated character recognition, achieving 94.9% accuracy on a 16,800-sample dataset they also released publicly, and has since been widely reused. Correspondingly, (Altwaijry & Al-Turaiki, 2020) extended this by evaluating CNNs on both AHCD and their newly introduced dataset called Hijja (children's handwriting) datasets, achieving 97% and 88% accuracy, respectively, highlighting the difficulty of recognizing children's handwriting. Similarly, (Ullah & Jamjoom, 2022) proposed a compact CNN architecture for handwritten Arabic characters, with 96.78% accuracy on AHCD. Although effective for character-level tasks, these models lack contextual modeling.

In (Balaha et al., 2021a, 2021b, 2021c), the authors introduced AHCR-DLS, a modular CNN system trained on the HMBD dataset, which includes contextual character forms and numerals. The model achieved 98.4% accuracy on cross-validation demonstrating strong scalability across datasets (CMATER, AIA9K).

In (Mostafa et al., 2021), they introduced OCFormer, a deep learning pipeline combining ResNet101 and Transformers for OCR of historical Arabic manuscripts, trained on a large-scale dataset of 30.5 million synthetic manuscripts lines (noisy, diacritic-rich images simulating real-world degradation). The model integrates Mask R-CNN for layout and line-level segmentation. However, due to computational limitations, only 15,000 samples were used in training, with evaluations on 2000-line subsets. The model achieved excellent performance in controlled settings (CER: 0.32%) and strong results under full-font and diacritic-rich conditions (CER: 7.27%, WER: 8.29%). Building on this, in (Mostafa et al., 2022), the authors proposed a fully end-to-end framework that replaces CNNs with BEiT-based Vision Transformers and a Transformer decoder, incorporating image enhancement, and post-correction module, achieving further gains in performance and scalability. Their enhanced model, trained on a 270-million-word corpus with 13 fonts and styles, achieved a CER of 4.46%, outperforming the previous model while reducing complexity. Together, these works advance Arabic OCR by addressing layout complexity, diacritics, and scalability.

The authors in (Eltay et al., 2022) addressed data imbalance using GAN-based augmentation, significantly boosting performance on low-resource classes in IFN/ENIT and AHDB. While (Nayef et al., 2022) introduced OLReLU, an improved activation function for CNNs, yielding 99% accuracy on AHCD, 95.35% on custom dataset, 91.23% on AIA9k, and 90% on Hijja dataset.

In (Momeni & BabaAli, 2022), the study compared a Transformer encoder–decoder with a Transformer Transducer on the KHATT dataset, showing superior sequence modeling without external language models.

In (Mehdi & Abdelghani, 2022), they implemented a Capsule Network for improved spatial feature learning in handwritten Arabic characters recognition. Their model integrates residual convolutional layers with dynamic routing mechanisms and was evaluated on IFHCDB, AHCD, and AIA9k achieving up to 97.57% accuracy.

DeepAHR, based on CNN, was introduced in (AlShehri, 2024). It consists of five convolutional layers and two fully connected layers, with intermediate activation, pooling, and batch normalization layers that enhance training stability and feature extraction. DeepAHR was evaluated on both the AHCD and Hijja datasets, achieving 98.66% and 88.24% accuracy, respectively. Its strong results, particularly

on the underrepresented Hijja dataset, makes it a valuable benchmark for style-diverse Arabic handwriting recognition.

Since the advent of deep learning, HATR has progressed from character-level models to end-to-end, context-aware systems. The field has moved from basic CNNs to hybrid CNN–RNN–CTC architectures, and now toward Transformer-based models, reflecting a shift toward segmentation-free recognition, reduced reliance on handcrafted features, and improved robustness across writing styles.

3.4.3 Hybrid Methods

Hybrid methods aim to leverage the complementary strengths of traditional and deep learning approaches to enhance recognition accuracy and system robustness. These systems often integrate handcrafted features with neural architectures or combine multiple deep learning components into a unified framework. While hybrid models address specific limitations of individual techniques, they typically involve more complex design, integration, and computational overhead.

In (Alma'adeed et al., 2004), they presented one of the earliest hybrid systems by integrating two recognition engines: the first employs a global feature framework with ascending and descending features coupled with a rule-based classifier, while the second incorporates a Hidden Markov Model (HMM) using an ensemble of stroke-based features. This dual-engine system captures both macro- and micro-level structural information from the word images.

While in (Elleuch et al., 2016), they proposed a CNN–SVM hybrid architecture for offline Arabic handwritten recognition. CNNs were used for hierarchical feature extraction, while an SVM classifier ensured better generalization through Structural Risk Minimization. The authors applied dropout to the fully connected layers to mitigate overfitting and used elastic deformation for data augmentation. When evaluated on HACDB and IFN/ENIT datasets, the system achieved 5.83% and 7.05% error rates, respectively, outperforming baseline CNN-only models and establishing a foundational approach for CNN–SVM integration.

In (Maalej & Kherallah, 2018), the authors developed a CNN–BLSTM–CTC model that enables segmentation–free Arabic word recognition. The CNN extracts spatial features, the BLSTM handles temporal dependencies, and the CTC loss allows for unsegmented label alignment. Applied to an extended IFN/ENIT dataset, the model achieved 92.21% accuracy. The combination of convolutional and sequential layers marked a transition toward end-to-end learning for Arabic handwriting, particularly in capturing context and handling character overlaps.

In (Doush et al., 2018), they introduced a post-processing model combining an error model and a contextual method for Arabic OCR. Trained on a 9000-image dataset, the model incorporated a unique alignment method called 'fast automatic hashing text alignment.' This work is notable as the first reported end-to-end post-processing system in Arabic OCR, addressing character-level correction using contextual cues.

In (Sobhi et al., 2020), they presented a CNN–RNN encoder–decoder model enhanced with a visual attention mechanism, trained on the KAFD dataset. The CNN encodes spatial features, the RNN models sequences, and the attention mechanism guides the decoder to focus on relevant image regions. The system achieved 89.82% accuracy, performing best on cleaner fonts. Despite not relying on language-specific rules, its use of RNNs may hinder scalability and real-time performance in heavily cursive or degraded texts.

Handwritten Arabic text segmentation integrating a combination of techniques, including a directed CNN with Mathematical Morphology Operations (MMO) is proposed in a successive work of (Elkhayati et al., 2020), (Elkhayati & Elkettani, 2021), and (Elkhayati et al., 2022). In (Elkhayati et al., 2020), the authors presented a multiagent approach, introducing an algorithm based on morphological analysis for line segmentation on Handwritten Arabic Proximity Datasets (HAPD) and KHATT databases. In (Elkhayati & Elkettani, 2021), they enhanced the approach by incorporating the External Features-based CNN (EFNet) to improve segmentation score on the same benchmark databases. Finally, in (Elkhayati et al., 2022), they focused on segmenting words into graphemes, addressing challenges like diacritics and overtraces, using a directed CNN and MMOs. The proposed strategy involves PD (Partial Dilation) and GE (Global Erosion) to reduce information loss. The directed CNN, focusing on targeted features, outperforms basic CNN, providing promising results. However, the authors acknowledge challenges related to vertical ligatures, isolated partial over-traces, and hidden ascenders, and expressed intentions to refine the approach by implementing techniques for these specific challenges.

In (Rahal et al., 2021), they proposed a hybrid system combining a deep sparse autoencoder (SAE) with Bag-of-Features (BoF) and HMMs. The method replaces conventional k-means clustering with SAE for dictionary learning, enabling the model to extract more discriminative features. Tested across P-KHATT, APTI, IFN/ENIT, and MNIST datasets, the model showed strong adaptability to multi-font and low-resolution inputs. While lacking modern deep sequence modeling, it offers an interpretable and resource-efficient alternative for OCR in challenging Arabic contexts.

Meanwhile, in (Alrobah & Albahli, 2021), they proposed a hybrid character recognition model that uses CNNs (HMB1 and HMB2 proposed in (Balaha et al., 2021a, 2021b, 2021c)) for feature extraction and SVM/XGBoost for classification. Evaluated on the Hijja dataset, the model reached

96.3% accuracy, with SVM yielding the best results. Their findings highlight the superior generalization of traditional classifiers over multilayer perceptron (MLP) in specific scenarios, especially under limited data conditions.

In (Balaha, Ali, Youssef, et al., 2021), they introduced HMB-AHCR-DLGA, a multi-strategy recognition framework combining deep learning, transfer learning, and genetic algorithms. Using the HMBD dataset, the study trained 14 CNN variants and incorporated pretrained networks (e.g., VGG16, MobileNetV2), with genetic algorithms used for hyperparameter optimization. The model achieved 92.88% accuracy and represents a scalable framework for architecture tuning and comparative benchmarking in HATR.

In (Aliwy & Al-Sadawi, 2021), the authors proposed a context-aware error correction module based on statistical language modeling. The method, trained on extensive corpora, identifies and rectifies both real-word and non-word errors. The system achieved up to 98% improvement, particularly in resolving errors related to image resolution, demonstrating the importance of linguistic post-processing in Arabic OCR pipelines.

In (Saidi et al., 2021), the study combined structural feature extraction with Radial Basis Function (RBF) neural networks. Their model focuses on features such as loops, diacritics, ascenders, and descenders from word skeletons. In parallel, (Bouchakour et al., 2021) introduced a model integrating Hu invariant moments, black pixel density, Gabor filters, and CNNs, coupled with histogram projection and contour-based segmentation. This combined method is well-suited for printed Arabic letter recognition but could be extended to handwritten cursive text with further development.

In (Ali & Mallaiah, 2022), they proposed a hybrid CNN–SVM system incorporating Spatial Pyramid Pooling (SPPnet) and Network-in-Network (NIN) modules to enhance spatial feature representation. The architecture also integrates dropout and ZCA whitening for regularization and introduces a Max-Margin Minimum Classification Error (M3CE) loss for better penalization of misclassifications. Evaluated on four datasets (AHDB, AHCD, HACDB, IFN/ENIT), the model reported up to 99% accuracy, particularly excelling in structured character-level recognition.

In parallel, (Lamtougui et al., 2022) developed a CNN–BLSTM–CTC pipeline that enables segmentation-free recognition of full Arabic word images. Combined with data augmentation strategies, their model improves generalization and sequence prediction without requiring pre-defined segmentation.

Similarly, (Gader & Echi, 2022) proposed a segmentation-free hybrid model integrating CNN, BLSTM, attention mechanisms, and CTC. Tested on IFN/ENIT and IAM datasets, the system achieved 97.1% accuracy and a 2.9% Character Error Rate (CER). This work] demonstrates strong cross-lingual generalization and offers an interpretable, high-performance architecture for HATR.

In (Maghraby & Samkari, 2023), they addressed the challenge of Arabic diacritic recognition by combining CNNs, Gated Recurrent Units (GRUs), and CTC loss. Trained on a small, manually annotated dataset of 110 word-level images with harakat, the model achieved a Character Recognition Rate (CRR) of 32.5% and a Word Recognition Rate (WRR) of 0%. While performance is constrained by dataset size, the study lays foundational work for diacritic-aware recognition systems in educational and religious contexts.

Lastly, (Aabed & Khairaldin, 2024) introduced a deep convolutional-BLSTM–CTC framework for full–line handwritten Arabic text recognition. Trained on the KHATT dataset, the system achieves 84% character-level and 71% word-level accuracy. Although it does not incorporate attention or transformer-based modules, its segmentation-free design offers practical advantages for document digitization, especially in historical and real-world use cases.

3.4.4 Comparative Analysis and Limitations

From this review, it is clear that deep learning approaches, particularly those leveraging CNNs and hybrid CNN–LSTM–CTC architectures, have established themselves as the state-of-the-art in HATR, demonstrating superior accuracy across benchmark datasets such as AHCD, Hijja, IFN/ENIT, and KHATT. CNNs excel in automatic feature extraction and spatial pattern recognition, while hybrid models effectively integrate spatial and sequential modeling through LSTMs and CTC loss, enabling segmentation-free, end-to-end recognition. However, these methods often require extensive labeled data and high computational resources, and they generally lack interpretability, limiting their deployment in low-resource or real-time settings.

Traditional methods, though largely outperformed in recent years, remain relevant due to their low complexity and interpretability, making them suitable for constrained environments or as baseline models. Hybrid techniques offer a middle ground, combining the strengths of deep and classical methods to improve generalization and robustness, particularly in complex or noisy inputs. Nevertheless, their architectural complexity and training requirements pose practical challenges. Emerging trends in transformer-based models promise enhanced contextual modeling and parallelization but face hurdles such as data scarcity and computational demands. Future research should prioritize the development of lightweight transformer variants for edge deployment, explore semi-supervised and multilingual training paradigms, and improve recognition of diacritic-rich scripts by incorporating semantic and contextual cues. A comparative summary of the three categories is presented in Table 4.

Table 4 Comparative summary of HATR methods

Characteristic	Traditional	Deep learning	Hybrid
Feature extraction	Handcrafted (zoning, chain code, projection)	Automatic via CNN, RNN, Transformers	Combined handcrafted + learned features
Model complexity	Low	High	Moderate to High
Performance	Moderate (limited to structured, clean input)	High (state-of-the-art on most benchmarks)	High (often better than standalone models)
Interpretability	High (transparent feature modeling)	Low (black-box models)	Moderate
Data requirement	Low to moderate	High (requires large labeled datasets)	Moderate to high
Scalability to full text	Poor (character-level only)	Excellent (word- and line-level recognition)	Varies (many still character-level)
Robustness to noise	Low	High (especially with augmentation)	High (with post-processing or ensemble strategies)
Ease of deployment	Easy	Challenging (due to size and hardware needs)	Varies depending on design
Popular datasets used	IFN/ENIT, APTI, AHCD	AHCD, KHATT, IFN/ENIT, Hijja, HMBD, AIA9K	Same as DL, often with custom augmentations
Related works	k-NN + zoning (Ali & Suresha, 2019b); HMM (Khorsheed, 2003)	CNN–BLSTM–CTC (Maalej & Kherallah, 2018) ViTs (Mostafa et al., 2022)	CNN–SVM (Elleuch et al., 2016); CNN–GRU–CTC (Maghraby & Samkari, 2023)

4 Databases

In the domain of HATR, access to standard and high-quality datasets is of utmost importance for research, development, and benchmarking of recognition systems (Al-Sheikh et al., 2020; Ramdan et al., 2013). Large, diverse dataset is crucial for training and evaluation of recognition models' performance effectively. Such datasets provide a wide range of examples and variations, enabling models to learn robust patterns and reliably adapt to unfamiliar and untrained data. Moreover, diverse datasets help in ensuring that the developed model is able to handle different conditions and scenarios, making it more reliable and accurate in real world applications. This section provides an overview of datasets commonly used for HATR models.

"IFN/ENIT (Institute for Communications Technology and Ecole Nationale d'Ingénieur de Tunis)" database is a collection of handwritten Arabic text presented by (Pechwitz et al., 2002). It comprises 26,459 samples of 946 town and village names in Tunisia, handwritten by 411 participants. This database was purposely intended for training and testing of HATR systems and is publicly available for research purpose at http://www.ifnenit.com/. However, the content vocabulary is limited (only contains names of cities in Tunisia).

AHDB database, created by Al-Ma'adeed et al. (2002) is a collection of Arabic words as well as sentences handwritten by one hundred distinct writers. It contains numbers written in words, twenty common Arabic words, and free Arabic text pages of writer's choice. The authors purposely created and made this dataset publicly available to offer training as well as evaluating datasets for research in this field. However, the dataset content is more suitable for applications such as cheque processing.

"Arabic Handwritten Text Database (AHTD)" developed by (Mahmoud et al., 2011), is comprised of two separate databases; ground-truth and images databases. That of images contains grey scale images of handwritten Arabic text that is divided into three sets of data: pages, paragraphs, and lines datasets, in which each set is accompanied by the ground-truth at the respective level. The other database contains ground truth data as well as the information of the writer that filled the respective forms, such as identification of the writer, age, gender, country, educational background, and their dominant writing hand (right- or left-handed). The database consists of 3000 pages and a total of 6000 paragraphs of Arabic text written by over 300 distinct writers. This database is applicable to Arabic handwriting related problems such as writer identification and text recognition.

"AHTID/MW (Arabic Handwritten Text Images Database/Multiple Writers) database" database (Mezghani et al., 2012) comprised of text lines and words images which totals to 3710 and 22,896, handwritten by 53 individuals of diverse ages as well as educational background. The authors have designed and made the database freely available.

HACDB (Handwritten Arabic Characters DataBase) was established by Lawgali et al. (Lawgali et al., 2013). The dataset comprises 6600 images of handwritten Arabic alphabets in their different forms and shapes (i.e., isolated, beginning, middle, end) handwritten by 50 participants between the age of 14–50. However, hamza, dots and diacritics have been removed leaving mainly the base characters, also ligatures are considered as single character. This is to mitigate the

complexities and the classes number. It is publicly available at http://computing.unn.ac.uk/characterdatabase/.

AHDB/FTR is an Arabic handwritten Text images database designed by (Ramdan et al., 2013), which consists of 497 image samples of cities' names in Libya handwritten by five Arabic scholars. This database is useful for problems related to the recognition of words, segmentation, or the task of identifying writers and it is freely available to researchers.

AIA9K (AlexU Isolated Alphabet) dataset, presented by (Torki et al., 2014), comprise of handwritten isolated Arabic characters that total's 8,737 images written by 107 participants between the ages 18–28 who are B.Sc. and M.Sc. students at Alexandria University.

KHATT (meaning 'handwriting' in Arabic, and also "KFUPM Handwritten Arabic TexT",), is a collection developed by (Mahmoud et al., 2014), which consist of Arabic text that is unconstrained, handwritten by a diverse set of 1000 different writers from various countries. It comprises a total of two thousand images of paragraph of the same text and another two thousand images of paragraph of dissimilar text, as well as their corresponding segmented text-line images. The database is publicly available for Arabic handwriting related research purposes at http://khatt.ideas2serve.net/index.php.

SUST-ALT ("Sudan University of Science and Technology-Arabic Language Technology group") (Musa, 2016) database is a compilation of 40,000 samples of handwritten Arabic digits, characters, names (both male and female), and Arabic online dataset written by 1000 participants. The dataset is freely available for research on Arabic handwriting related problems at www.sustech.edu.

DBAHCL, is an Arabic handwritten alphabets and ligatures database that was developed by (Lamghari & Raghay, 2017). It holds 15,400 image samples of which 5500 are samples of the different forms of the Arabic characters and 9900 are samples of ligatures, handwritten by 50 writers. The database can be accessed by researchers at http://sa-sd.net/dbahcl.

Hijja is a dataset of handwritten Arabic characters handwritten by children of ages 7–12 crea'ted by (Altwaijry & Al-Turaiki, 2020). It is also the first dataset designed specifically for recognizing children's handwritten Arabic characters. It consists a total of 47,434 samples of Arabic characters in their different forms, handwritten by 591 participants. This dataset is made available at https://github.com/israksu/Hijja2.

DBAHD ("DataBase for Arabic Handwritten Diacritics") presented by (Lamghari & Raghay, 2021), is a database that includes 5 forms of Arabic handwritten diacritical markings (one, two, and three dots, Hamza, and Maddah). It comprises a total of 500 sample images of the five forms of handwritten Arabic diacritics with 100 samples for each. This database is made available at https://github.com/nidaliso/DBAHD/.

HMBD dataset introduced by (Balaha et al., 2021a, 2021b, 2021c), consist of 54,115 grayscale images of handwritten Arabic characters and numerals, collected from 125 participants aged 9 to 82. It contains 115 classes altogether, of which 105 classes represent handwritten Arabic characters in their contextual forms, and 10 classes represent handwritten Arabic numerals, isolated in separate folders. While HMBD is one of the comprehensive publicly available datasets for handwritten Arabic character recognition, it excludes diacritical marks, and lacks complete positional forms coverage, particularly in medial and final contexts. It can be found at the following address https://github.com/hossambalaha/hmbd-v1.

Muharaf, introduced by (Saeed et al., 2024), is a large-scale, annotated dataset comprising 1644 handwritten Arabic manuscript pages from the 19th to twenty-first centuries. It captures real-world variability in handwriting, quality, and degradation. It includes over 36,000 annotated text lines, spatial polygons, and layout metadata, supporting handwriting recognition, layout analysis, and writer identification. Muharaf stands out for its diversity and annotation quality, filling a critical gap in historical Arabic document digitization research. Details on the dataset can be found at https://github.com/MehreenMehreen/ScribeArabic.

Although online handwriting datasets such as ADAB are valuable resources for evaluating real-time recognition systems, they fall outside the primary scope of this review. Our analysis centers on offline HATR, and thus the datasets discussed are limited to those predominantly used in offline HATR research and are publicly available. A summary of the datasets is provided in Table 5.

5 Performance Evaluation Metrics

Performance evaluation of Arabic text recognition models is important for assessing their accuracy, efficiency and robustness. It is used to identify the strengths and weaknesses of recognition models, which leads to continuous improvement and advancement in the field. This typically includes assessing the accuracy of the identified Arabic text compared to the input image text and consequently measuring the recognition quality by computing different performance metrics (Faizullah et al., 2023). In many cases, the performance of a recognition system relies on the input image quality, however, investigating performance rates such as recognition rate, error rate, and rejection rate is common practice (Eikvil, 1993).

1. Recognition rate, also known as accuracy, is the percentage of correctly recognized words or characters in the output in comparison to the actual labels. The metric is obtained by calculating the ratio of correctly recognized

Table 5 Summary of datasets commonly used in handwritten Arabic text recognition (HATR)

Dataset	Year	Number of samples	Participants	Content						Availability
				Characters	Words	Text	Numbers	Diacritics	Ligatures	
IFN/ENIT (Pechwitz et al., 2002)	2002	26,459	411 (Tunisian)	✗	✓	✗	✗	✗	✗	Public
AHDB (Al-Ma'adeed et al., 2002)	2002	–	100	✗	✓	✗	✓	✗	✗	Public
AHTD (Mahmoud et al., 2011)	2011	–	300+(International)	✗	✗	✓	✗	✗	✗	Public
AHTID/MW (Mezghani et al., 2012)	2012	3,710	53 (Arabic Natives)	✗	✗	✓	✗	✗	✗	Public
HACDB (Lawgali et al., 2013)	2013	6,600	50 (aged 14–50)	✓	✗	✗	✗	✗	✗	Public
AHDB/FTR (Ramdan et al., 2013)	2013	497	5 (Libyan)	✗	✗	✓	✗	✗	✗	Public
AIA9K (Torki et al., 2014)	2014	8,737	107 (Students aged 18–25)	✓	✗	✗	✗	✗	✗	Public
KHATT (Mahmoud et al., 2014)	2014	–	1000 (International)	✗	✗	✓	✗	✗	✗	Public
SUST-ALT (Musa, 2016)	2016	40,000	1000 (Sudanese)	✓	✓	✗	✓	✗	✗	Public
DBAHCL (Lamghari & Raghay, 2017)	2017	15,400	50	✓	✗	✗	✗	✗	✓	Public
Hijja (Altwaijry & Al-Turaiki, 2020)	2020	47,434	591 (children aged 7–12)	✓	✗	✗	✗	✗	✗	Public
DBAHD (Lamghari & Raghay, 2021)	2021	500	–	✗	✗	✗	✗	✓	✗	Public
HMBD (Balaha et al., 2021a, 2021b, 2021c)	2021	54,115	125	✓	✗	✗	✗	✗	✗	Public
Muharaf (Saeed et al., 2024)	2024	36,000	–	✗	✗	✓	✗			Public

characters or words and that of total count of characters or words in the ground truth.

2. Error rate measures the percentage of incorrectly recognized characters or words compared to the ground truth. It is computed at different granularities based on the edit distance such as Levenshtein distance, which quantifies the minimum number of operations required to transform the recognized text into the ground truth. Lower edit distance indicates higher similarity between the recognized text and ground truth (Najam & Faizullah, 2023).
   - Character Error Rate (CER): is the measure of edit distance that is between the recognized characters and the ground truth, considering substitutions, insertions, and deletion of characters.
   - Word Error Rate (WER): is the measure of edit distance that is between the recognized words and that of the ground truth, also considering substitutions, insertions, and deletions of words.
3. Rejection rate computes the percentage of characters or words the system fails to recognize. In the context of Arabic text recognition, this indicates the frequency of instances the model is uncertain about the correct transcription due to low confidence scores or ambiguous input.

Other commonly used metrics to assess the effectiveness of HATR models include precision, which computes the ratio of instances correctly recognized and total number of predicted instances as positive, recall, which calculates the ratio of instances correctly recognized and total number of actual instances that are positive, and precision, harmonic mean, F1-Score, and recall, which offers a balanced measure of model performance.

6 Challenging Factors

HATR models face numerous challenging factors which hinders accurate and efficient extraction of text from images or scanned documents. Such challenges stem from various sources including the following:

1. Arabic script complexity:
   - Intricate features such as ligatures, diacritics, and contextual variations such as shown in Table 1 and Fig. 4 pose challenges for recognition systems.
   - Complex ligatures formed by adjacent characters require specialized handling to accurately recognize individual components.

- Vowel diacritics and other diacritics are in many cases considered noise and therefore often omitted in handwritten text, which leads to ambiguity in recognition.

2. Variability in Handwriting styles (refer Fig. 8) makes it quite challenging for recognition models.
3. Limited Annotated Datasets:
   - Annotated datasets for handwritten Arabic text are particularly scarce compared to other languages, which makes it challenging to train and evaluate recognition models.
   - Existing datasets often lack diversity in terms of writing styles and text sizes, which can limit the generalization capability of models.
4. Noisy input data, including low-quality scans degraded documents, and background noise can adversely affect the performance of HATR models, especially in real-world scenarios.
5. Computational resources constraints:
   - Deep learning models often require substantial computational resources for complex computational tasks such as segmentation, feature extraction, and sequence modelling, leading to high computational cost and slower processing speeds.

7 Recent Trends

The field of HATR has seen notable advancements in recent years, driven by developments in deep learning and natural language processing (NLP). Emerging research trends include:

1. End-to-end, segmentation-free models: that eliminate traditional preprocessing pipelines enabling more streamlined and accurate recognition (Gader & Echi, 2022; Maghraby & Samkari, 2023), (Aabed & Khairaldin, 2024; Lamtougui et al., 2022; Mostafa et al., 2022).

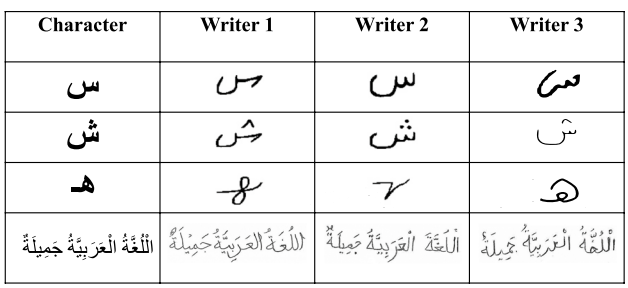


Fig. 8 Variation in handwriting styles

2. Transformer-based architectures: including lightweight ViT variants, which facilitate global context modeling and improved scalability (Mostafa et al., 2022).
3. Multilingual and cross-domain HATR systems: capable of generalizing across different scripts and dialects.
4. Integration of attention mechanisms: that enable models to focus on relevant regions of the input image, enhancing recognition accuracy (Gader & Echi, 2022; Sobhi et al., 2020).
5. Transfer Learning and Pre-training: where models pre-trained on large corpora are fine-tuned on HATR datasets, improving performance, particularly in low-resource scenarios (Awni et al., 2022; Balaha, Ali, Youssef, et al., 2021).
6. Real-time and mobile-ready HATR systems, reflecting growing demand for deployable applications in education, documentation, and translation.
7. Strategies for low-resource settings: where recent work explores augmentation techniques and synthetic data generation using GANs to enrich class diversity and address data scarcity (Eltay et al., 2022).

These trends reflect the field's trajectory toward more robust, scalable, and user-friendly recognition systems that can operate across diverse contexts and constraints.

8 Discussion and Conclusion

This review has provided a comprehensive synthesis of the methods, challenges, and trends in Handwritten Arabic Text Recognition (HATR). By tracing the progression from traditional handcrafted feature-based methods to hybrid and deep learning techniques, we highlighted how recent models, particularly end-to-end and Transformer-based architectures, have reshaped the field.

Key insights include:

1. The effectiveness of segmentation-free and attention-based models in addressing the complexities of cursive, diacritic-rich Arabic text.
2. The growing role of transfer learning and multimodal integration in improving model robustness with limited annotated data.
3. Persistent limitations in dataset availability, computational resource demands, and model interpretability.

Although each reviewed approach has demonstrated effectiveness, performance depends heavily on the specific task requirements, dataset characteristics, and available resources. Strategic integration of multiple methods and fine-tuning of pre-trained models, especially Transformers, often yields superior recognition accuracy. Despite

these advancements, major challenges remain, particularly in recognizing diacritics and ligatures, handling overlapping characters, and developing resource-efficient models. Addressing these issues requires innovative algorithms for segmentation and feature extraction, as well as robust handling of noisy or degraded inputs. Furthermore, building comprehensive and diverse datasets that capture a wide range of writing styles and diacritical variations remains critical. Augmentation techniques should also consider the linguistic characteristics and nuances of Arabic handwriting. Ongoing research in deep learning, transfer learning, and multimodal processing presents promising avenues for advancing the state of the art. This review identifies open research gaps, especially in contextual modeling and data scarcity, that must be addressed to build more inclusive and intelligent HATR systems.

Future directions include integrating HATR with other modalities such as speech and named entity recognition (NER), leveraging large language models (LLMs), supporting various Arabic dialects, and designing resource-efficient architectures. These innovations will be essential for deploying robust, real-world HATR systems across diverse applications and platforms.

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Data Availability Statement This review mentions several datasets relevant to the field of Arabic text recognition. Publicly available datasets include IFN/ENIT, HACDB, KHATT, SUST-ALT, DBAHCL, and DBAHD, which can be accessed via http://www.ifnenit.com/; http://computing.unn.ac.uk/characterdatabase; http://khatt.ideas2serve.net/index.php; www.sustech.edu; http://sa-sd.net/dbahcl; and https://github.com/nidaliso/DBAHD respectively. Researchers interested in accessing datasets, such as AHDB; AHTD; AHTID/MW; AHDB/FTR; AIA9K; and Hijja, may contact the authors of the original study detailed in (Al-Ma'adeed et al., 2002; Altwaijry & Al-Turaiki, 2020; Mahmoud et al., 2011; Mezghani et al., 2012; Ramdan et al., 2013; Torki et al., 2014) respectively. No new datasets were generated or analyzed in this study.

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